MIL-B-9978A(MU)
7 September 1967
SUPERSEDING
MIL-T-9978
24 July 1963

MILITARY SPECIFICATION

BIOLOGICAL AGENT, SIMULANT,

BACILLUS GLOBIGII: DISPENSER, AN-MI

1. SCOPE

1.1 This specification covers one type of simulant agent dispenser system for use in the release of a bacterial aerosol of Bacillus globigii.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-C-310 - Chlorofluoro Hydrocarbons (of the Methane and Ethane Series).

FSC 1380

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PPP-B-636 - Box, Fiberboard.

PPP-T-60 - Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging.

PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-L-19896 - Labels and Label Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant.

MIL-S-50003 - Simulant Agent, BG-I, Bacillus Globigii Spores, Dried.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

MUNITIONS COMMAND

D100006 - Biological Agent, Simulant BG-I; Dispenser, AN-MI.

(Copies of specifications, standards and drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C., 20025. Orders for this publication should cite "49 CFR 71-90".)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois, 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D100006 shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.1.2 Components. All components of the dispenser shall conform to the specifications listed on Drawing D100006, or to the specific configuration shown on Drawing D100006.

3.2 Assembly. The dispenser shall be assembled as specified on Drawing D100006.

3.3 Composition of the charge. The composition of the charge in the dispenser shall conform to the requirements given in table I when tested as specified in the test paragraph referenced therein.

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Table I. - Composition of the Charge

Composition	Specification	Requirements	Test Paragraph
Simulant Agent, BG-I, Bacillus Globigii Spores, Dried	MIL-S-50003	No less than 2.25 ± 0.25 gm	4.6.1
Chlorofluoro Hydrocarbons (of the Methane and Ethane Series)	BB-C-310 (Type 12)	300 + 6.0 gm	4.6.1

3.4 Moisture content. The maximum moisture content of the charge shall be 0.04 percent when tested as specified in 4.6.2.

3.5 Leakage. The dispenser shall not leak when tested as specified in 4.6.3.

3.6 Functioning. The dispenser shall discharge an aerosol when tested as specified in 4.6.4.

3.7 Heating. The dispenser shall show no evidence of leakage, distortion, or other defects due to heating when tested as specified in 4.6.5.

3.8 Nozzle discharge pattern. The nozzle shall have a uniform discharge pattern and direction when tested as specified in 4.6.6.

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3.9 Marking. The marking on the dispenser shall be lithographed, embossed, roller coated, stamped, or labelled in white letters. If labels are used, they shall be type IV, color blue No. 35109, MIL-L-19896. Marking shall be in accordance with MIL-STD-129 and shall read as follows:

BIOLOGICAL AGENT,

SIMULANT, BACILLUS GLOBIGII:

DISPENSER, AN-MI

FSN- 1380-025-3274

Specification: MIL-B-9978

Contract No:

FOR CHEMICAL-BIOLOGICAL-RADIOLOGICAL

TRAINING AND FIELD USE

INGREDIENTS

BACILLUS GLOBIGII - 2 grams

DICHLORODIFLUOROMETHANE - 300 grams

SUPPLIER's NAME ICC SYMBOL - 2P

ADDRESS

DATE FILLED: (month and year) Gross Wt. _____

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DIRECTION FOR USE

HOLD DISPENSER UPRIGHT AND SHAKE VIGOROUSLY. REMOVE PROTECTIVE CAP AND PRESS VALVE BUTTON. IF SPRAY DOES NOT OPERATE OR IF DISCHARGE IS SMALL, REMOVE AND CLEAN THE VALVE BUTTON. (CLEANING MAY BE ACCOMPLISHED BY PASSING A SMALL WIRE OR SIMILAR ITEM THROUGH THE VALVE BUTTON OR BY USING COMPRESSED AIR.) REPLACE THE VALVE BUTTON AND REPEAT THE ABOVE INSTRUCTIONS.

PERISHABLE: DO NOT STORE IN TEMPERATURES OVER 100°F.

3.10 Bacillus globigii. The bacillus globigii shall conform to MIL-S-50003 at the time of filling. (See 6.5.)

3.11 Storage of Bacillus globigii. The bacillus globigii shall be stored in a sealed container at a temperature not to exceed 80°F.

3.12 Preproduction. When specified in the contract order, a preproduction sample of dispensers shall be produced in accordance with this specification for examination and tests. (See 6.2.)

3.13 Workmanship. The workmanship shall be of the quality necessary to produce filled dispensers free from defects which would affect the proper functioning and serviceability. The dispenser shall be free from contamination (dirt, oil, grease or other foreign matter).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Special provisions.

4.2.1 Alternative inspection procedures. The supplier may utilize any alternative inspection procedures or inspection equipment which will insure equal or better quality assurance by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.10, 3.11, and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the dispensers produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of bacillus globigii shall be included in any one lot of filled dispensers.

4.4 Sampling.

4.4.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 For test. Sampling for the tests specified in 4.6.1, 4.6.2, 4.6.4 and 4.6.6 shall be conducted in accordance with MIL-STD-105, level SI, with an AQL of 2.5 percent defective. For the test specified in 4.6.5, two dispensers out of 500, or a fraction thereof, shall be taken.

4.5 Examination. The sample dispensers, sample components, and level A and level B preparation for delivery shall be examined in accordance with the classification of defects and MIL-STD-105.

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4.5.1 Classification of defects.

4.5.1.1 Biological agent, simulant, BG-1: dispenser, AN-M1 (Dwg D100006).

Categories	Defects	Acceptance standard
Critical:	None defined	
Major:	AQL 0.65 percent defective	
101	Leakage	4.6.3
Major:	AQL 1.5 percent defective	
102	Component missing or incorrectly assembled	
103	Diptube length incorrect	
104	Cap does not fit snugly on can	
105	Valve sealing cap dimple missing	
106	Nozzle outlet dia (.018) incorrect	
107	Nozzle outlet dia (.040) incorrect	
108	Nozzle outlet thickness (.038) incorrect	
109	Workmanship (3.13)	
Major:	AQL 2.5 percent defective	
110	Nozzle discharge pattern	4.6.6

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4.5.1.2 Preparation for delivery (see section 5).

Categories	Defects	Acceptance standard
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Quantity in box incorrect	
102	Cap missing or loose	
103	Fiberboard box incorrect	
104	Partitions missing or incorrect	
105	Liner missing or incorrect	
106	Closure incorrect	
107	Strapping missing or incorrect	
108	Tape missing or incorrect	
109	Marking missing or incorrect	

4.6 Tests. Tests shall be conducted as follows.

4.6.1 Composition of the charge.

4.6.1.1 Procedure. Weigh the sample dispenser and record the weight in grams (gm) as W_1. Cool the dispenser so that its contents are at -25°F (a bath of dry ice and alcohol may be used). After cooling, punch an approximate 3/8-inch hole in the dispenser above the liquid level of its contents. Under a hood, gradually allow the temperature of the dispenser to increase so that the type 12 gas is released at a uniform but not a rapid violent rate. Recool if necessary to control the release rate. After discharge of the type 12 gas has ceased, dry the dispenser at 50°C for approximately one hour. Allow the dispenser to cool to room temperature and weigh. Record the weight of the dispenser in gm as W_2. Thoroughly rinse the bacillus globigii from the container by using distilled water. Make a final rinse using alcohol. Dry the empty container to a constant weight and record the final weight in gm as W_3.

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4.6.1.2 Calculation.

$W_1 - W_2$ = weight of the type 12 gas

$W_2 - W_3$ = weight of the simulant agent

where: W_1 = weight of the dispenser

W_2 = weight of the container and simulant agent

W_3 = weight of the empty container

Compare the weights obtained for compliance with the requirements specified in 3.3.

4.6.2 Moisture content.

4.6.2.1 Blank determination. Connect the sampling device (see figure 1 and 6.6) to a Fisher-Milligan or equivalent gas washing bottle by means of a rubber tube. Aspirate air, dried over phosphorus pentoxide (P_2O_5), or an equivalent drying agent, through the apparatus for several hours to remove moisture. Introduce approximately 100 milliliters (ml) of absolute methanol into the gas washing bottle. Titrate the methanol with Karl Fischer reagent (see 6.4) to the red end point and add 30.0 ml of the reagent in excess. Disconnect the sampling device from the rubber tubing and aspirate approximately 300 gm of dry type 12 gas, conforming to BB-C-310, through the gas washing bottle at a rate that droplets will not be carried to the top of the bottle. Titrate the excess Karl Fischer reagent with a standard water-in-alcohol solution.

4.6.2.2 Blank value.

$30.0 - AB = C$

where: A = ml of standard water-in-alcohol solution.

B = titer value of the standard water-in-alcohol solution with respect to the Karl Fischer agent.

C = blank value.

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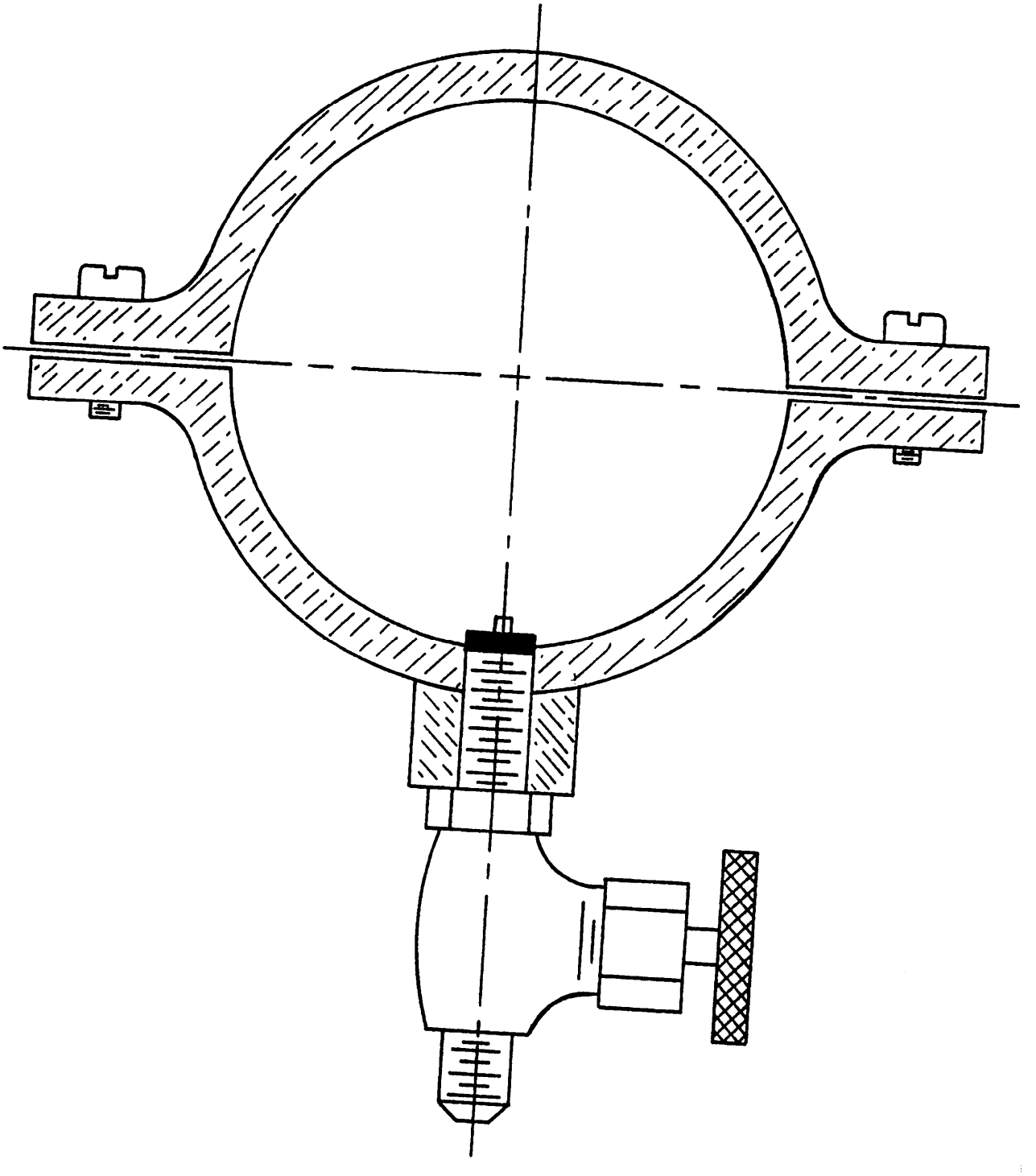


Figure 1 - Detail of Sampling Device

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4.6.2.3 Moisture determination. Accurately weigh the sample dispenser recording the weight in gm as W_1. Place the dried sampling device with a rubber gasket over the needle around the dispenser adjacent to the bottom seam. Tightly close the valve on the sampling device. Tighten the $\frac{1}{2}$ inch bolts on the sampling device until the needle penetrates the dispenser and the rubber gasket is compressed sufficiently to make a gas-tight seal. Introduce approximately 100 ml of absolute methanol into the dry gas washing bottle, titrate with Karl Fischer reagent to the red end point, and add 30.0 ml of reagent in excess. Invert the dispenser and support it in this position. Connect the sampling device to the gas washing bottle and slowly open the valve on the sampling device to allow the gas to bubble through the solution at a rate that droplets are not carried to the top of the bottle. When discharge of the gas has ceased, tightly close the valve on the sampling device and disconnect it from the gas washing bottle. By a suitable method to prevent absorption of moisture, transfer the methanol-reagent solution from the gas washing bottle into the dispenser, seal, and thoroughly agitate the solution in the dispenser. Allow the dispenser to stand for 5 to 10 minutes, transfer the solution from the dispenser into a dry container, protecting the solution from absorption of moisture. Titrate the excess reagent with a standard water-in-alcohol solution to the first disappearance of the red color. Record the volume of the standard water-in-alcohol solution as A. Thoroughly rinse the dispenser with absolute methanol, and discard the rinse. Place the empty dispenser in an oven and dry at a temperature of $80^{\circ} \pm 5^{\circ}C$. When the dispenser is dry, cool to room temperature, weigh and record the weight in gm as W_2.

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4.6.2.4 Calculation.

$$\frac{(30.0 - AB - C)\ D}{W\ (1000)} \times 100 = M.$$

where: A = volume in ml of the standard water-in-alcohol solution.

B = titer value of the standard water-in-alcohol solution with respect to the Karl Fischer reagent.

C = blank value.

D = milligrams of water equivalent to one (1) ml of Karl Fischer reagent.

W_1 = weight in gm of dispenser.

W_2 = weight in gm of empty dispenser.

$W_1 - W_2 = W$ - weight in gm of charge.

M = percent moisture in sample.

4.6.3 Leakage. Establish an equilibrium temperature to adjust the contents of the dispenser to 137.5° ± 2.5°F. Submerge the dispenser for 2 ± 0.25 minutes, approximately two (2) inches below the surface of water maintained at 137.5° ± 2.5°F and containing a wetting agent. Observe the dispenser for leakage which will be indicated by a continuous emission of bubbles.

4.6.4 Functioning. Vigorously shake the dispenser as stated in the directions for use, and operate the valve momentarily. Observe the dispenser for discharge of an aerosol.

4.6.5 Heating. Heat the sample dispenser for no less than one (1) hour in a water bath maintained at a temperature of 130° ± 5°F, and observe for evidence of leakage or distortion.

4.6.6 Nozzle discharge pattern. Prepare the target by brushing a mixture of three parts talc and one part crystal violet dye (See 6.3) on to filter paper of suitable size. Install nozzle assembly in a can charged with 300 gm of a mixture consisting of 50 percent Freon 12 and 50 percent ethyl alcohol by volume. Place the target 12 ± 1 inches from the nozzle. Release a short burst of the aerosol from the nozzle and observe the pattern of the discharge for a general uniform minimum six (6) inch circular configuration on the target.

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4.7 Acceptance/rejection criteria.

4.7.1 Heating. If any sample dispenser tested in accordance with 4.6.5 fails to comply with the heating requirement specified in 3.7, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Twenty-four dispensers shall be packed in a V5c, style SFF-L, fiberboard box conforming to PPP-B-636 in one layer of four (4) rows with six (6) units each. Slotted fiberboard partitions made of material conforming to W5c of PPP-B-636 shall form individual cells for each dispenser. The box shall be provided with a liner having vertical corrugations of the same material as the partitions. The liner shall be of one piece, and the ends thereof shall abut in the middle of one side of the box. The box shall be strapped. Closure and strapping shall be in conformance with the appendix to PPP-B-636. In addition, all seams including the manufacturer's joint shall be sealed with a minimum 3-inch wide, type III, class 1 pressure sensitive tape conforming to PPP-T-60; the tape shall extend over all corners and edges at least 3 inches.

5.1.2 Level B. The dispensers shall be packed the same as level A (see 5.1.1) except that a type I, class 1, fiberboard box (275-pound test board) conforming to PPP-B-636 and gummed paper tape conforming to PPP-T-76 shall be used.

5.1.3 Level C. A uniform number of dispensers shall be packed in a manner to insure carrier acceptance and safe delivery at first destination. Shipping containers shall comply with the Uniform Freight Classification Ratings, Rules, and Regulations or regulations of other carriers applicable to the mode of transportation. Unless otherwise specified by the procuring activity, the shipping containers shall be of uniform size, shape, and material.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with MIL-STD-129, ICC Regulations 49 CFR 71-90 for dichlorodifluoromethane, and with the following:

PERISHABLE: DO NOT STORE AT TEMPERATURES OVER 100°F

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6. NOTES

6.1 Intended use. The dispenser covered by this specification is for use in Chemical-Biological-Radiological field use and training.

6.2 Ordering data. The procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

(c) Preproduction requirements (when required).

1. Quantity required.

2. Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

3. Name and address of test facility when testing is performed by the Government.

4. Shipping instructions when applicable.

5. Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Dye. The dye should be Crystal Violet, Extra Pure (DuPont). Care should be taken to keep away from water which also reacts with the dye.

6.4 Karl Fischer reagent. Karl Fischer reagent may be prepared and standardized as specified in ASTM Standard D1123.

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6.5 Bacillus globigii spores. The following suggestions applicable to bacillus globigii spores are offered for suppliers engaged in filling dispensers in accordance with this specification:

(a) This material is hygroscopic and adequate precautions should be taken against introducing moisture into it during filling operations.

(b) The organism is not generally toxic to man, however, some persons may become allergic to it. Therefore, a respirator should be worn when handling or working with the organism.

(c) The organism is very persistent and can quickly contaminate an area. Adequate controls to prevent contamination should be taken.

(d) A solution of calcium hypochlorite and water containing 5 percent available chlorine is recommended as a decontaminating agent.

(e) Further information pertinent to decontamination may be obtained from the contracting officer.

6.6 Sampling device. Further details on the sampling device shown in figure 1 are shown on Edgewood Arsenal Drawing C136-301-29, available upon request from the contracting officer.

Custodian:

Army - MU

Preparing activity:

Army - MU(FD)

Project No. 1380-A054

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